

EXPERIMENT 7B

DATA VISUALIZATION USING TABLEAU

Aim:

To connect, analyze, visualize, and create interactive dashboards using Tableau for effective business intelligence and data analytics.

Algorithm: Transaction Data Visualization using Tableau

Step 1: Data Loading

Connect the Transaction dataset to Tableau.

Verify the fields: Transaction ID, Timestamp, Sender Name, Amount (INR), and Status.

Step 2: Transaction Amount Bar Chart

Create a worksheet for transaction amounts.

Place Sender Name on Columns.

Place Amount (INR) on Rows.

Select the Bar Chart visualization.

Enable data labels to display the transaction amounts.

Step 3: Transaction Status Pie Chart

Create a worksheet using Status and Transaction Count.

Count the transactions for each status.

Select the Pie Chart visualization.

Display SUCCESS and FAILED transactions with labels.

Step 4: Transaction Trend Line Chart

Create a Line Chart using Timestamp and transaction amount.

Place Timestamp on Columns.

Place Amount (INR) on Rows.

Select the Line Chart visualization.

Use the chart to observe transaction trends over time.

Step 5: Calculated Field

Create a calculated field named Transaction Amount.

Use the formula:

SUM([Amount (INR)])

Display the calculated transaction amount in the worksheet.

Step 6: Status Filter

Add Status to the Filters shelf.

Select SUCCESS or FAILED.

Observe how the charts change according to the selected status.

Step 7: Transaction Parameter

Create a parameter named Select Measure.

Add the options Transaction Amount and Transaction Count.
Use the parameter to switch between transaction amount and transaction count in the visualization.

Step 8: Dashboard Creation

Create a new Tableau Dashboard.

Add the Transaction Amount Bar Chart, Transaction Status Pie Chart, and Transaction Trend Line Chart.

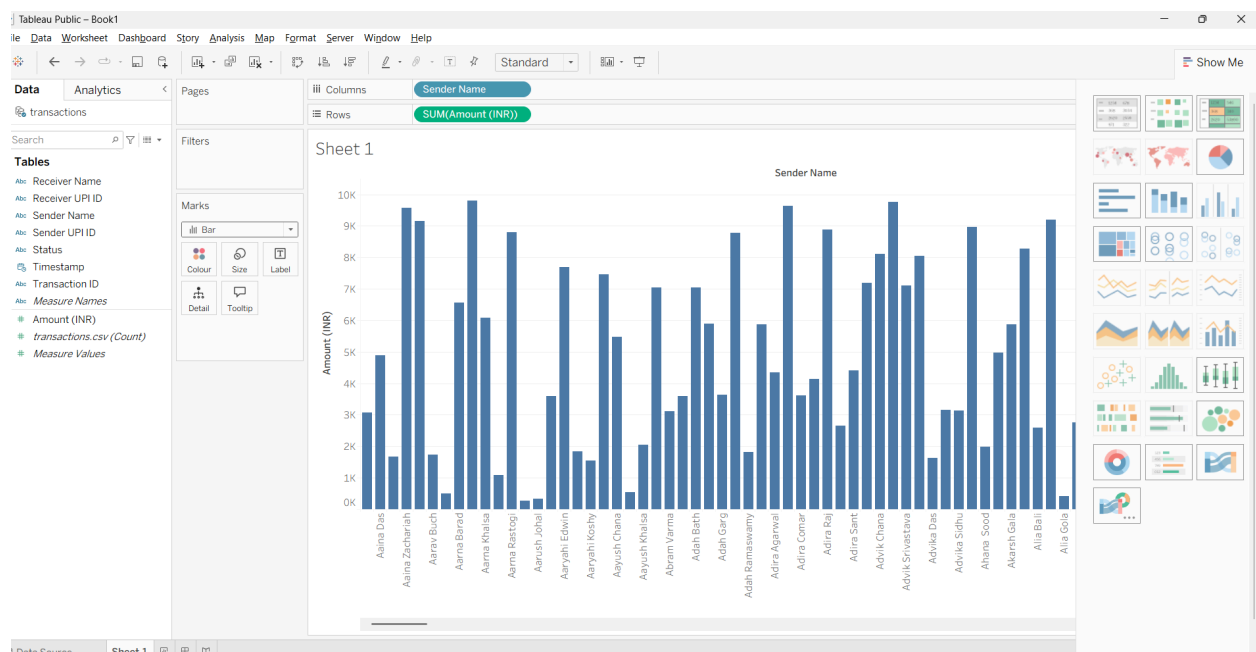
Add the Status filter and Select Measure parameter.

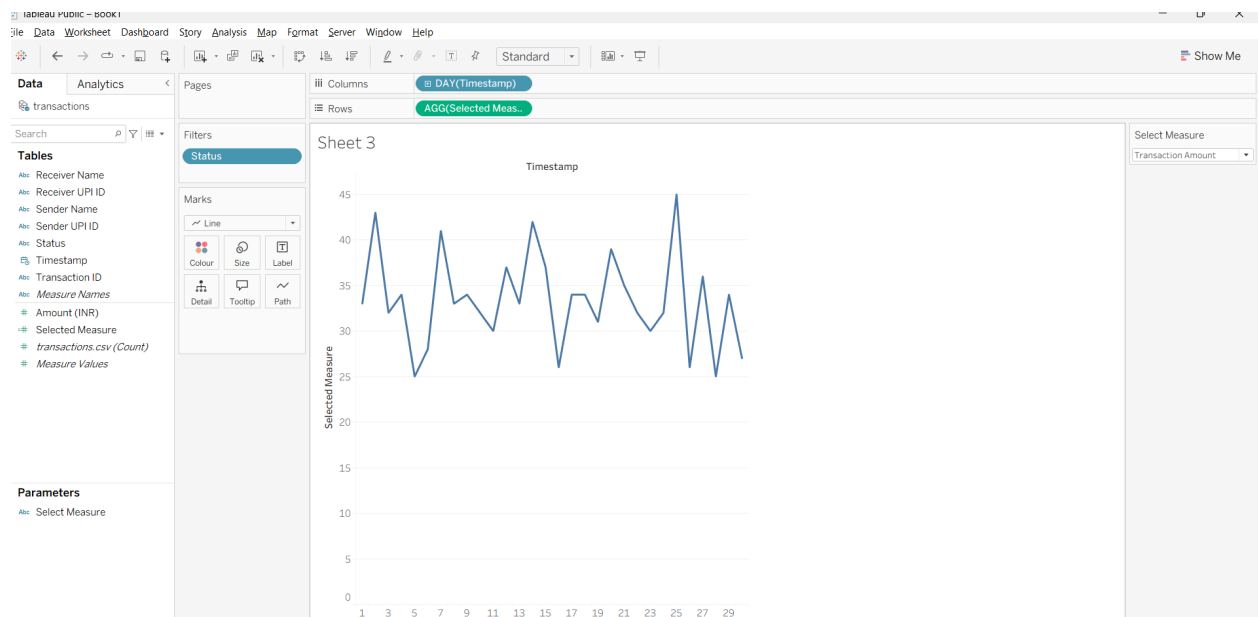
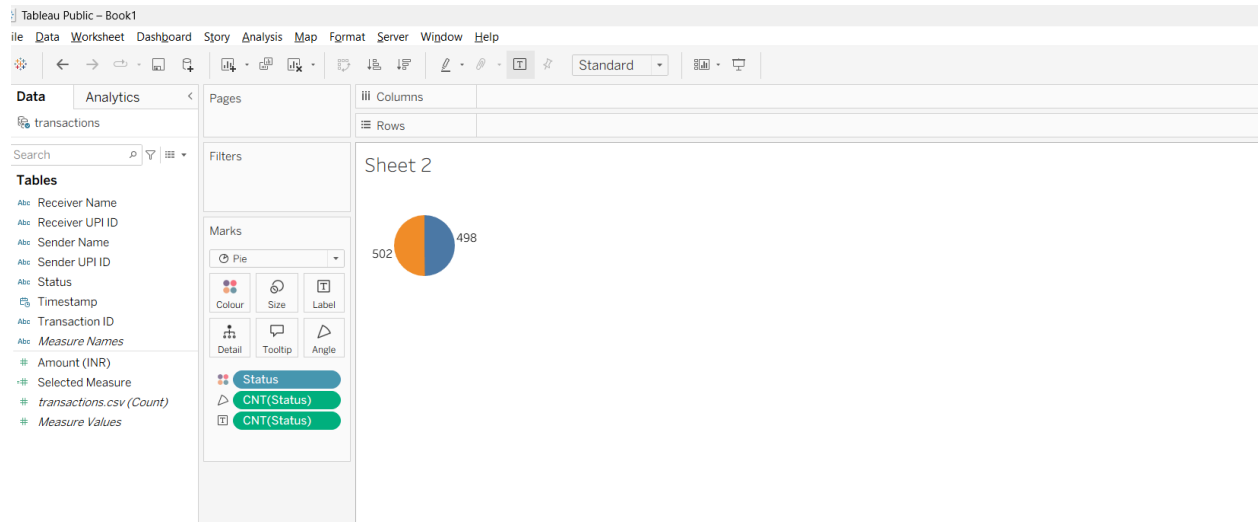
Arrange the visualizations clearly.

Step 9: Analysis

Analyze the charts to identify the senders with the highest transaction amounts, the number of SUCCESS and FAILED transactions, changes in transaction activity over time, and the effect of the Status filter and Select Measure parameter.

Output:





Result:

The transaction dataset was successfully visualized in Tableau using bar charts, pie charts, line charts, calculated fields, filters, and parameters.